



Department of Software Engineering

University of Engineering and Technology, Taxila

SDA

(LAB)

Project Report

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Instructor:

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SmartSalon Management System

1. Introduction

SmartSalon Management System is a desktop-based application developed using Java 21 and JavaFX 21. The system is designed to automate salon operations including customer management, appointment booking, staff management, invoicing, feedback handling, and business reporting. The project follows Object-Oriented Programming (OOP) principles and implements multiple software design patterns to ensure scalability, maintainability, and code reusability.

2. Objectives

The main objectives of the project are:

- To digitize salon management operations.
- To provide customers with an easy appointment booking system.
- To assist administrators in managing services, staff, customers, and invoices.
- To generate reports and analytics for business decision making.
- To demonstrate the implementation of software design patterns in a real-world application.

3. Technologies Used

Technology	Purpose
Java 21	Core Programming Language
JavaFX 21	Graphical User Interface
Maven	Project Build Management
Java Serialization	Data Storage
CSS	User Interface Styling
FXML	UI Layout Design

4. System Architecture

The application follows the MVC (Model-View-Controller) architecture.

Model

Contains business entities such as:

- User
- Customer
- Admin
- Staff
- Appointment
- Invoice
- Feedback
- Notification

View

Contains JavaFX FXML files and CSS styling for the user interface.

Controller

Handles user interactions and communication between the View and Model layers.

5. Design Patterns Implemented

5.1 Factory Pattern

The Factory Pattern is implemented through the ServiceFactory class.

Purpose:

- Creates service objects dynamically.
- Reduces object creation complexity.
- Improves maintainability.

Example Services:

- Haircut Service
 - Facial Service
 - Hair Coloring Service
-

5.2 Decorator Pattern

The Decorator Pattern allows additional salon services to be added dynamically.

Examples:

- Hair Styling Add-on
- Massage Add-on
- Premium Treatment Add-on

Benefits:

- Extends functionality without modifying existing code.
 - Supports flexible service customization.
-

5.3 Observer Pattern

The Observer Pattern is used for notification management.

Whenever an appointment status changes:

- Customers receive notifications.
- Administrators receive updates.

Classes Used:

- AppointmentSubject
 - AppointmentObserver
 - CustomerNotifier
 - AdminNotifier
-

5.4 Singleton Pattern

Singleton ensures only one object instance exists throughout the application.

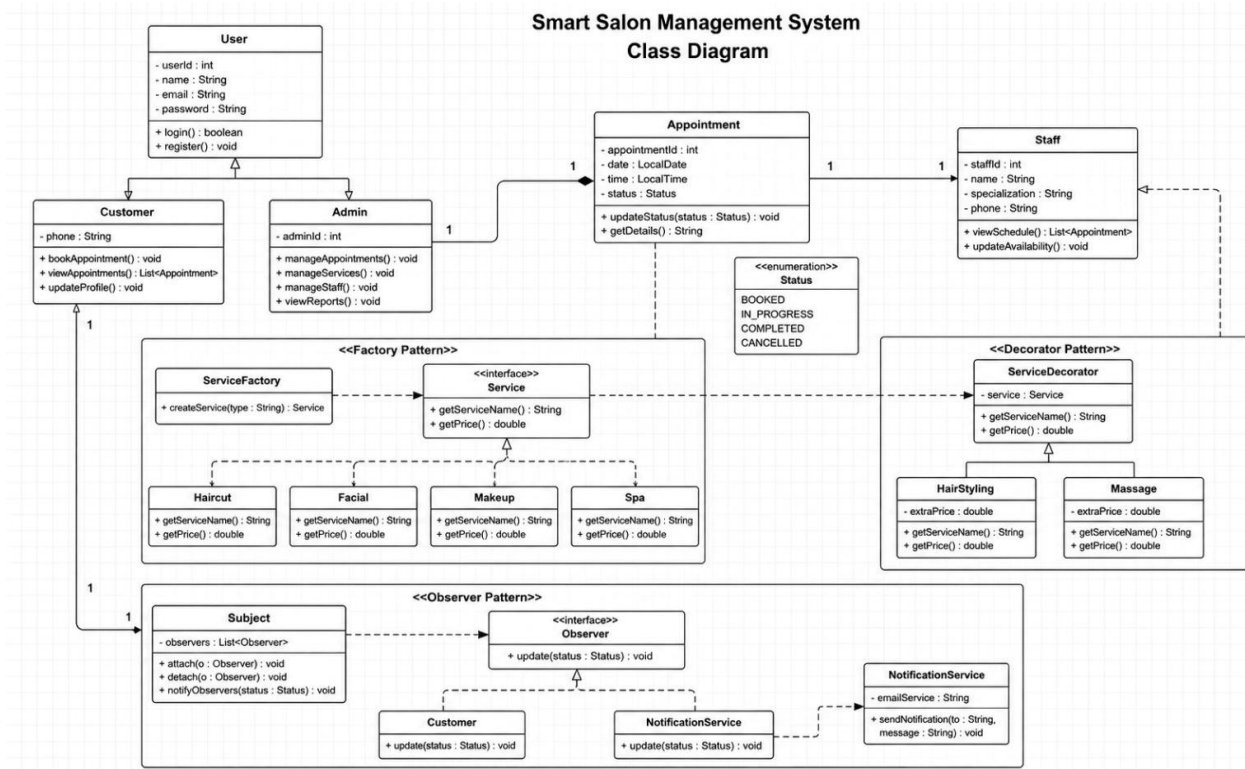
Classes:

- SessionManager
- NotificationPublisher

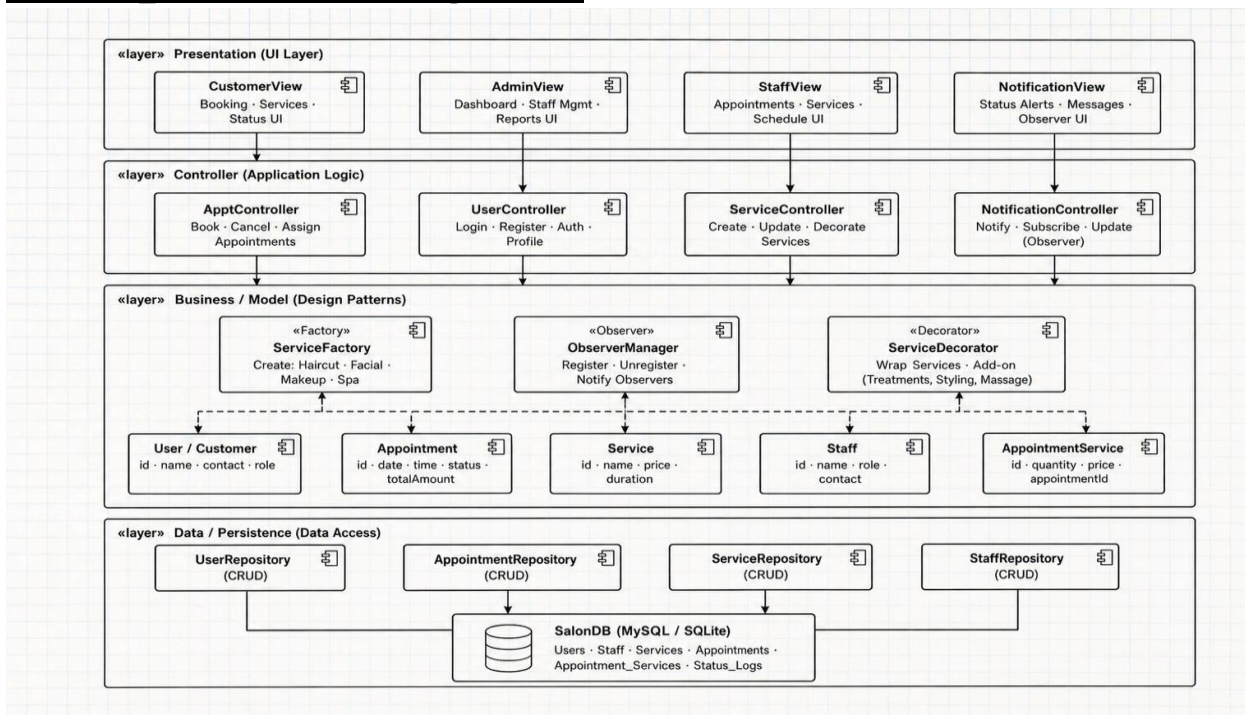
Benefits:

- Centralized session management.
 - Efficient memory utilization.
-

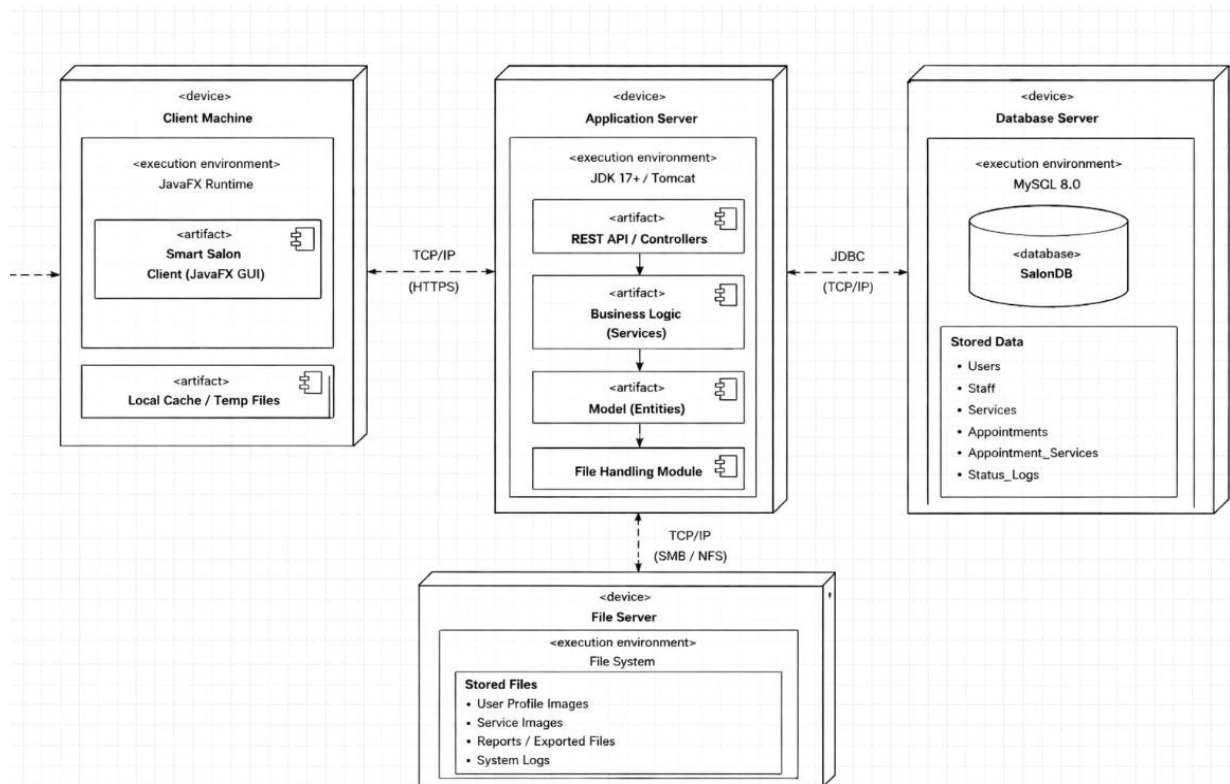
6. System Diagrams: Class Diagram



Component Diagram

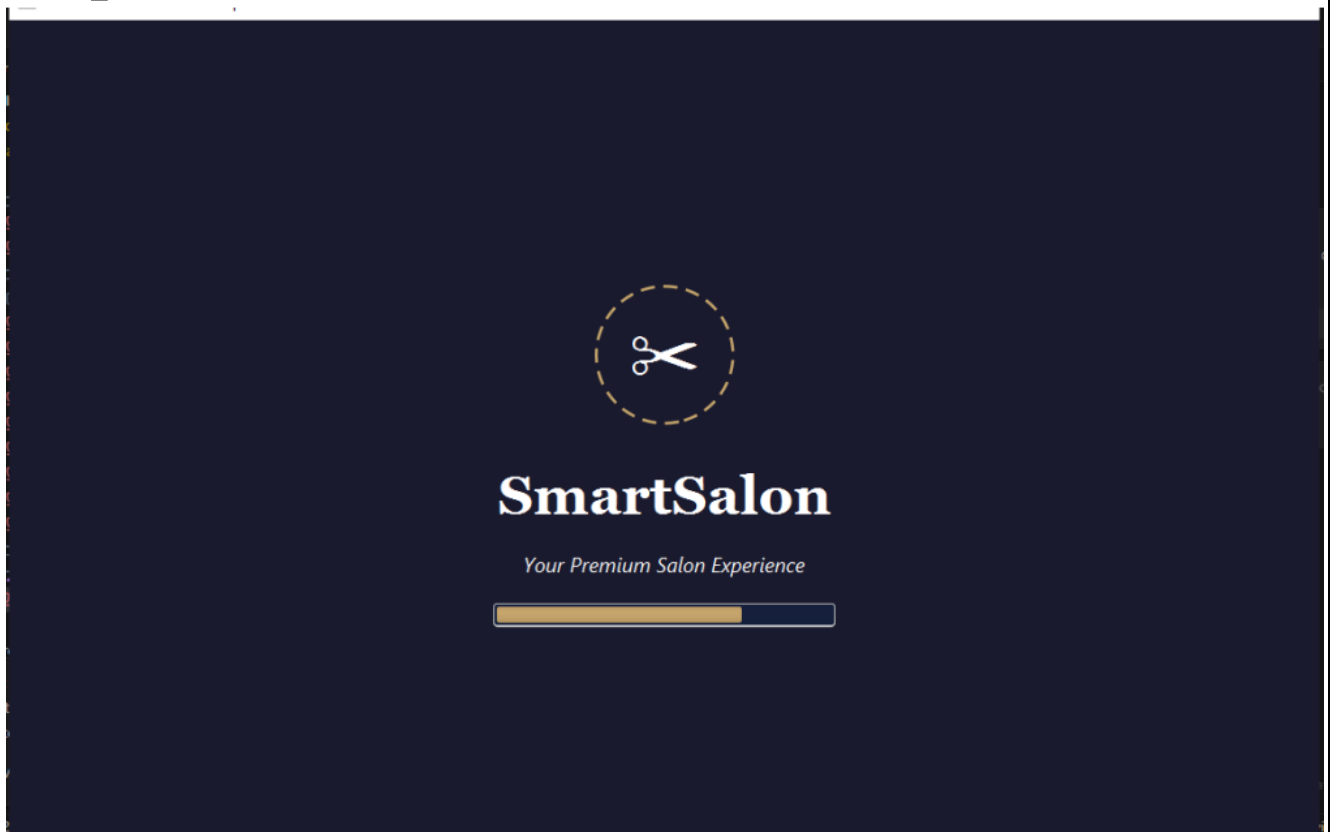


Deployment Diagram

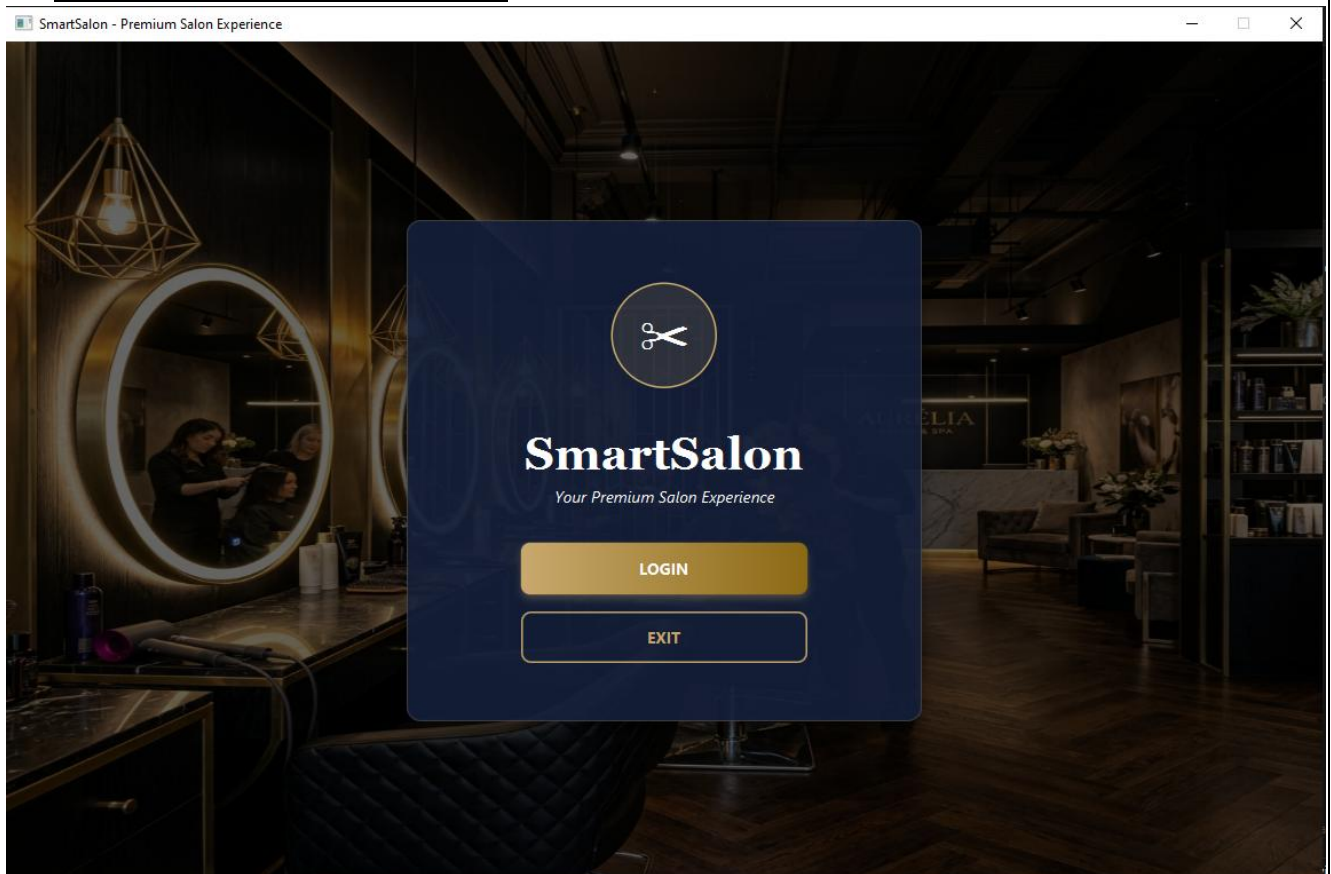


SMART SALON MANAGEMENT SYSTEM

- Splash Screen

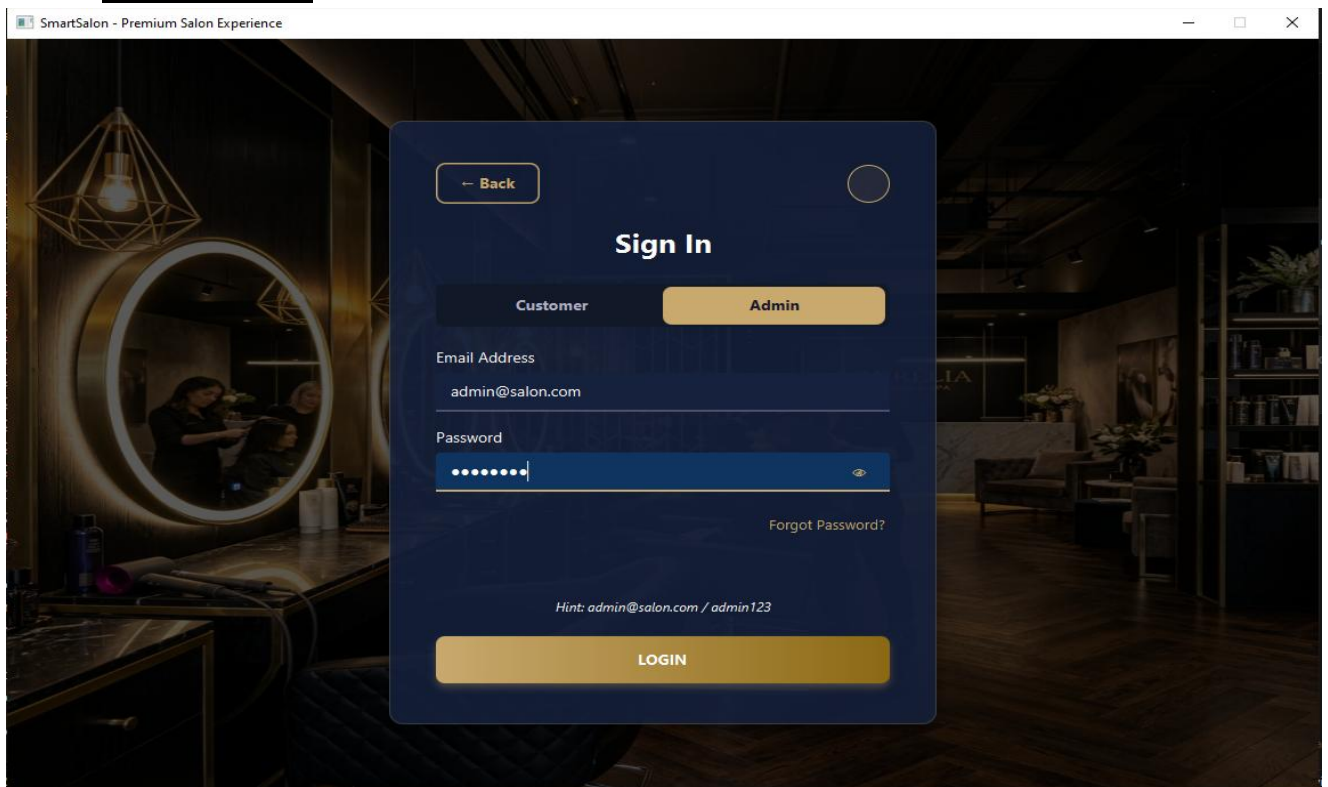


- Welcome Screen

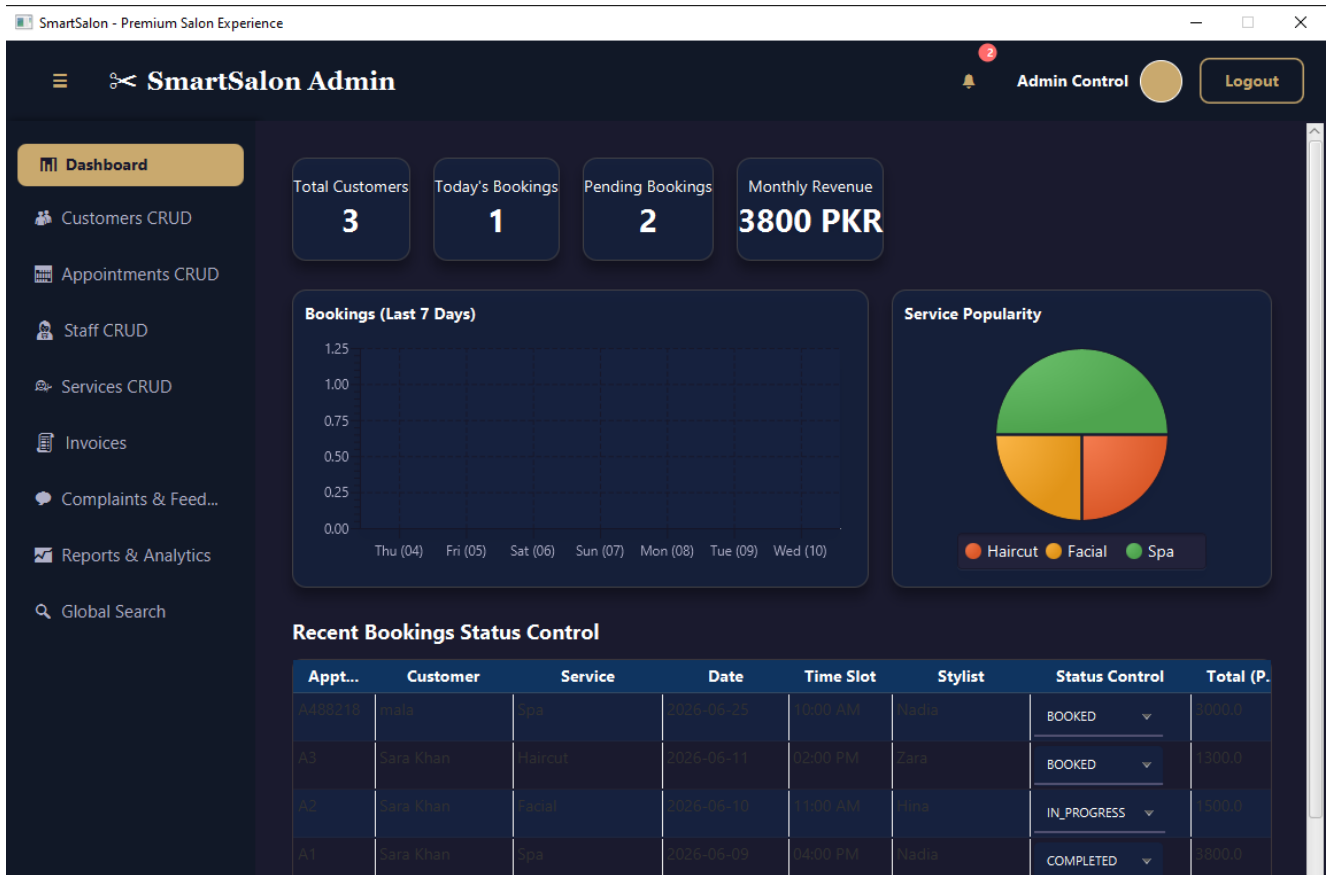


• Login Screen

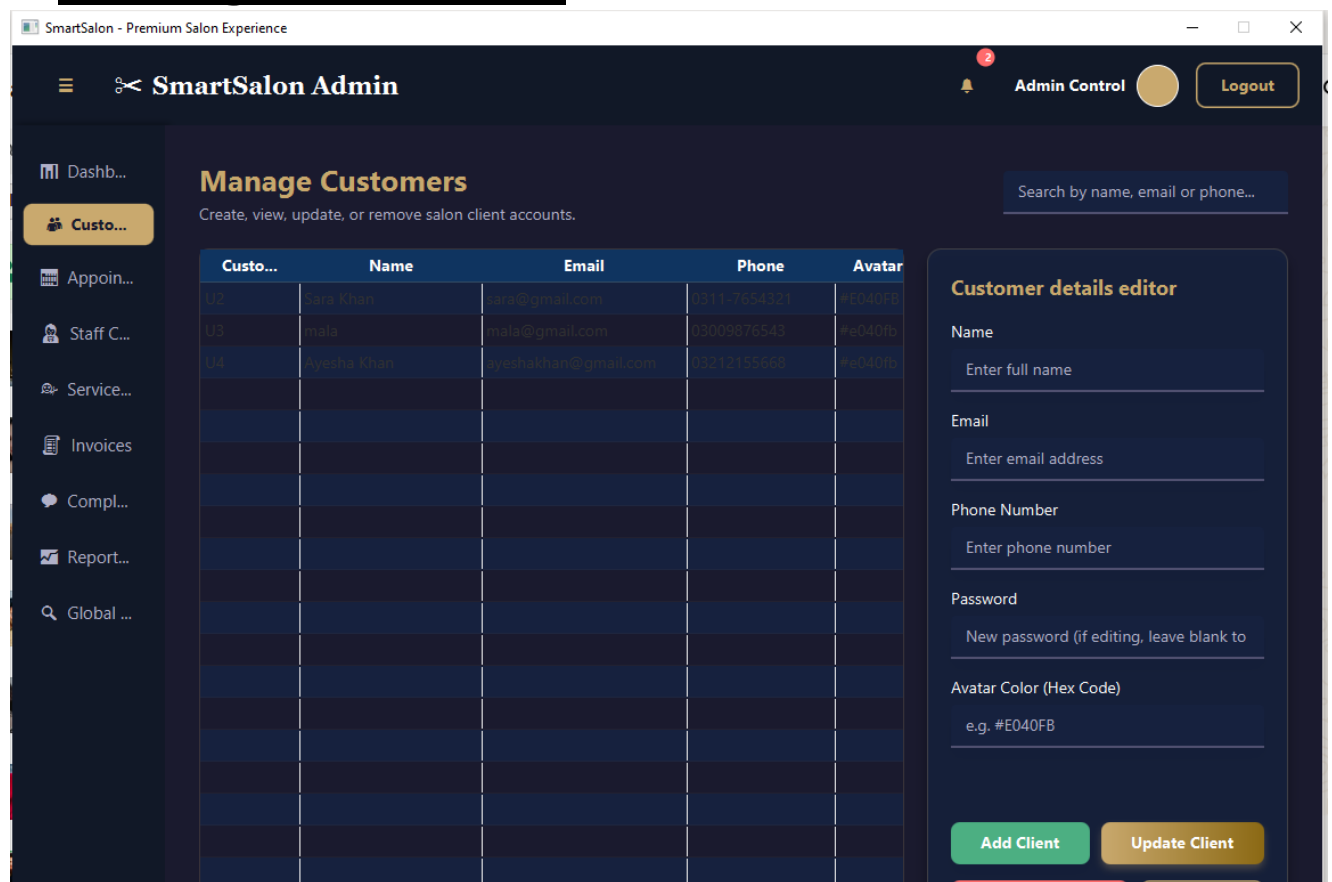
Admin:



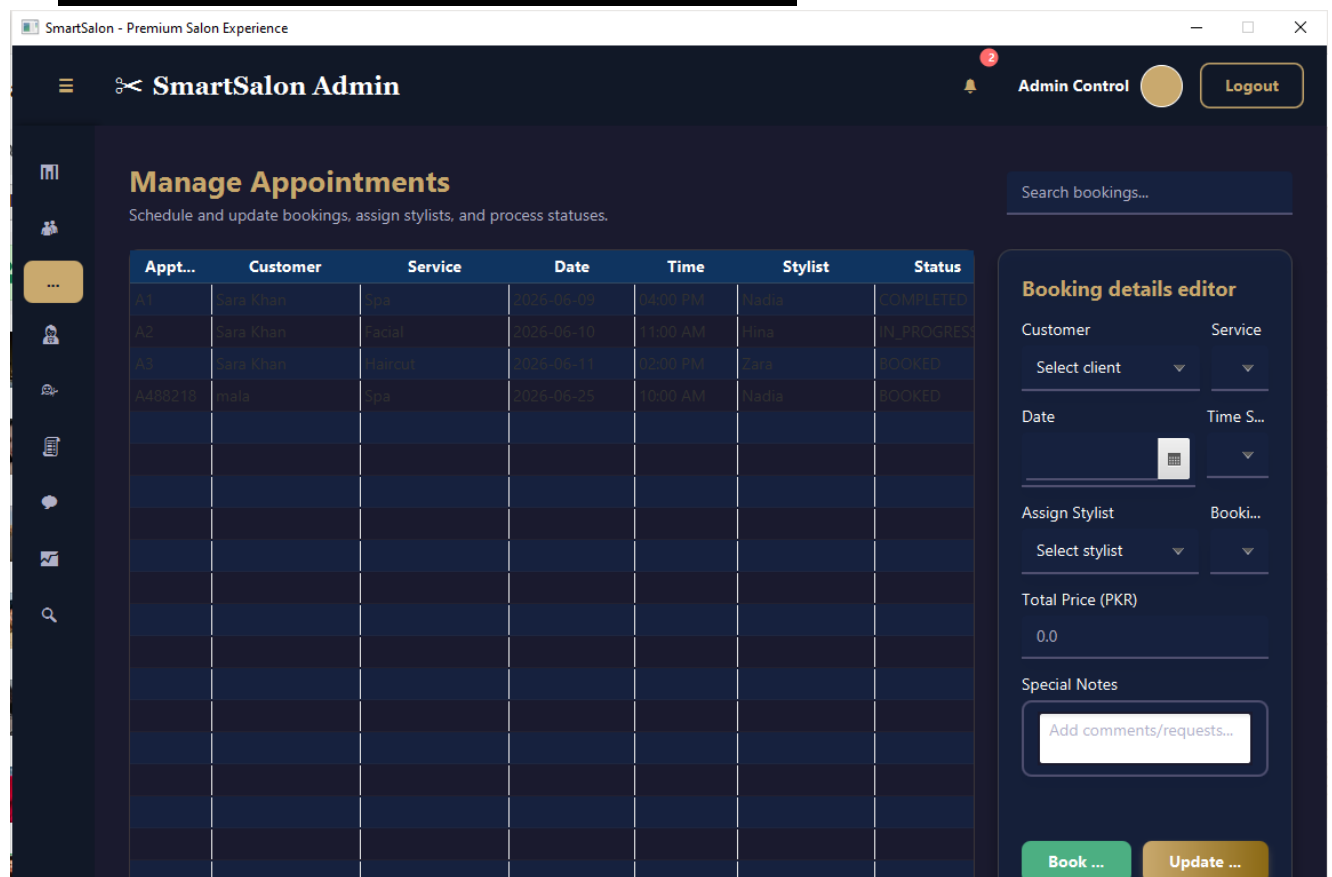
Admin Dashboard



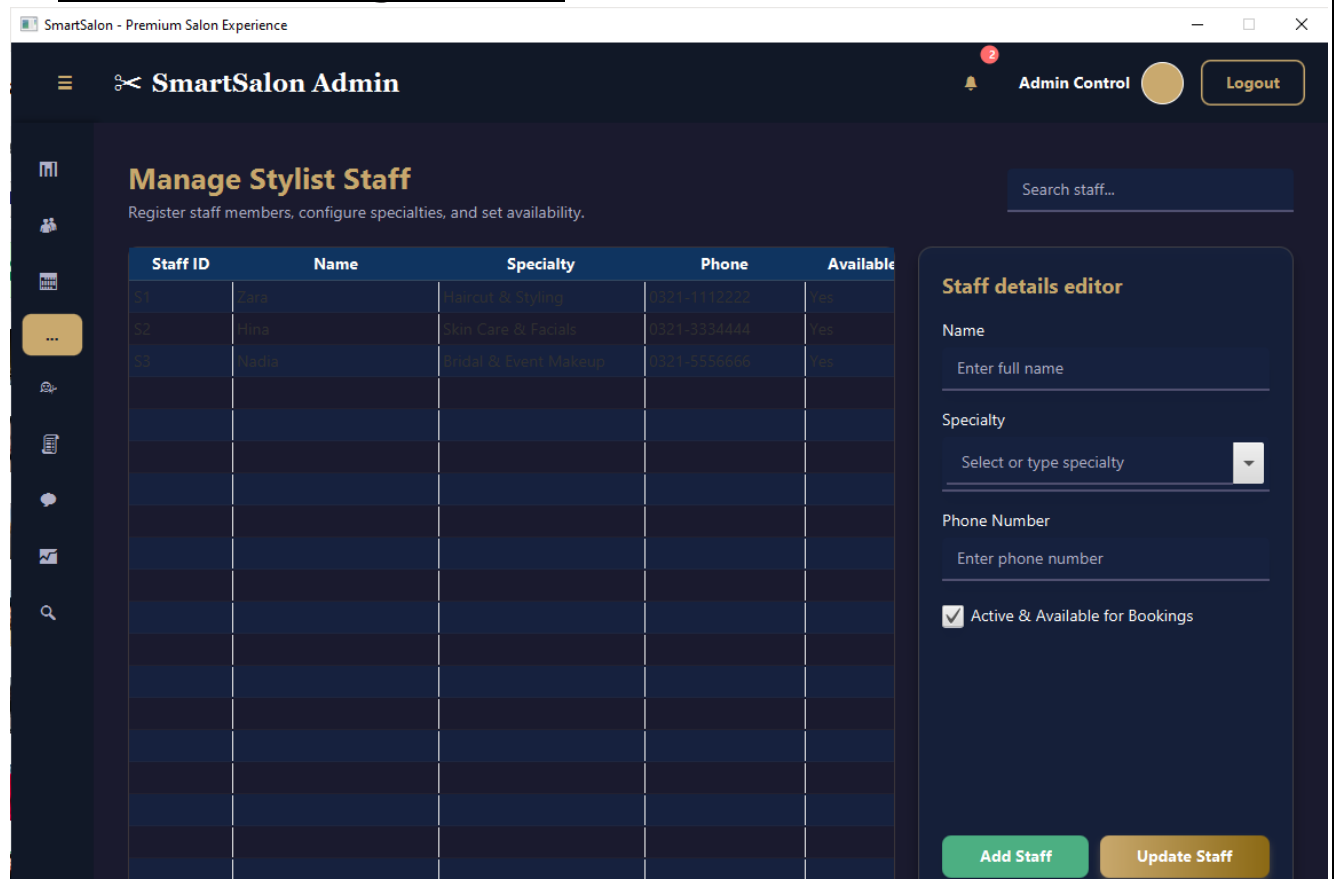
- **Manage Customer**



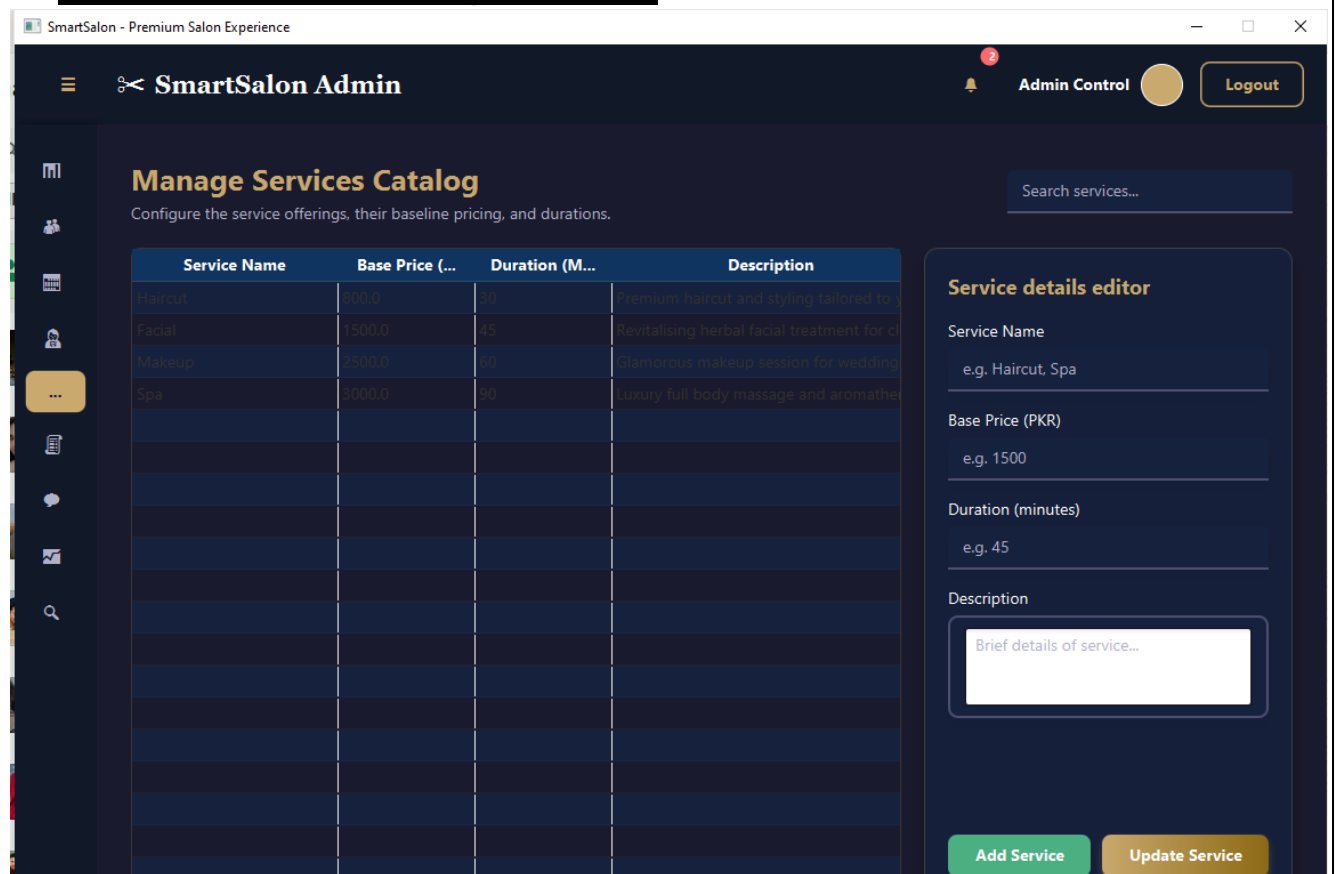
- **Appointment Management**



- **Staff Management**



- **Services Management**



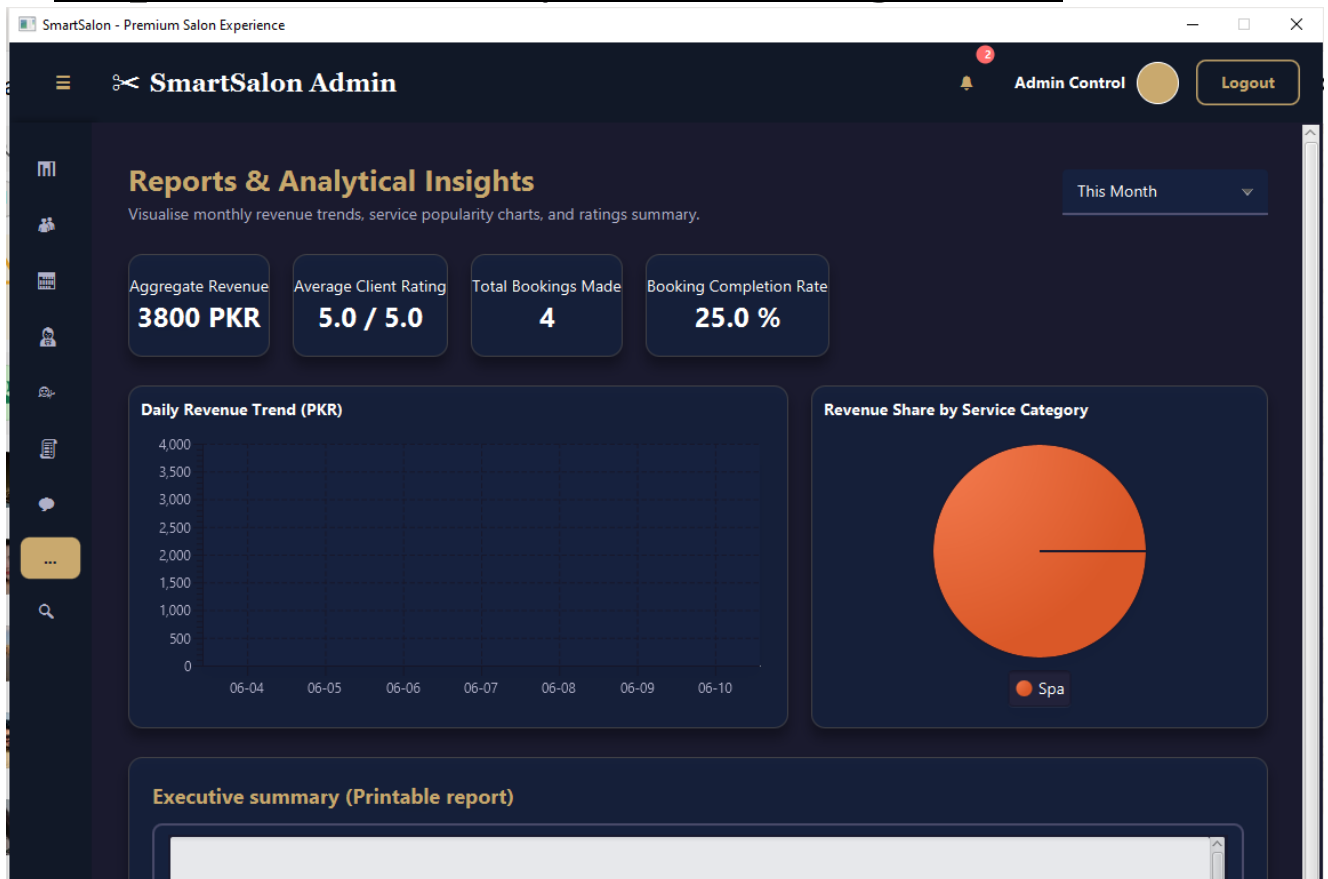
- **Invoice and Billing Management**

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• Feedbacks and Complaints Management

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• Reports and Analytical Management



• Global Database Search

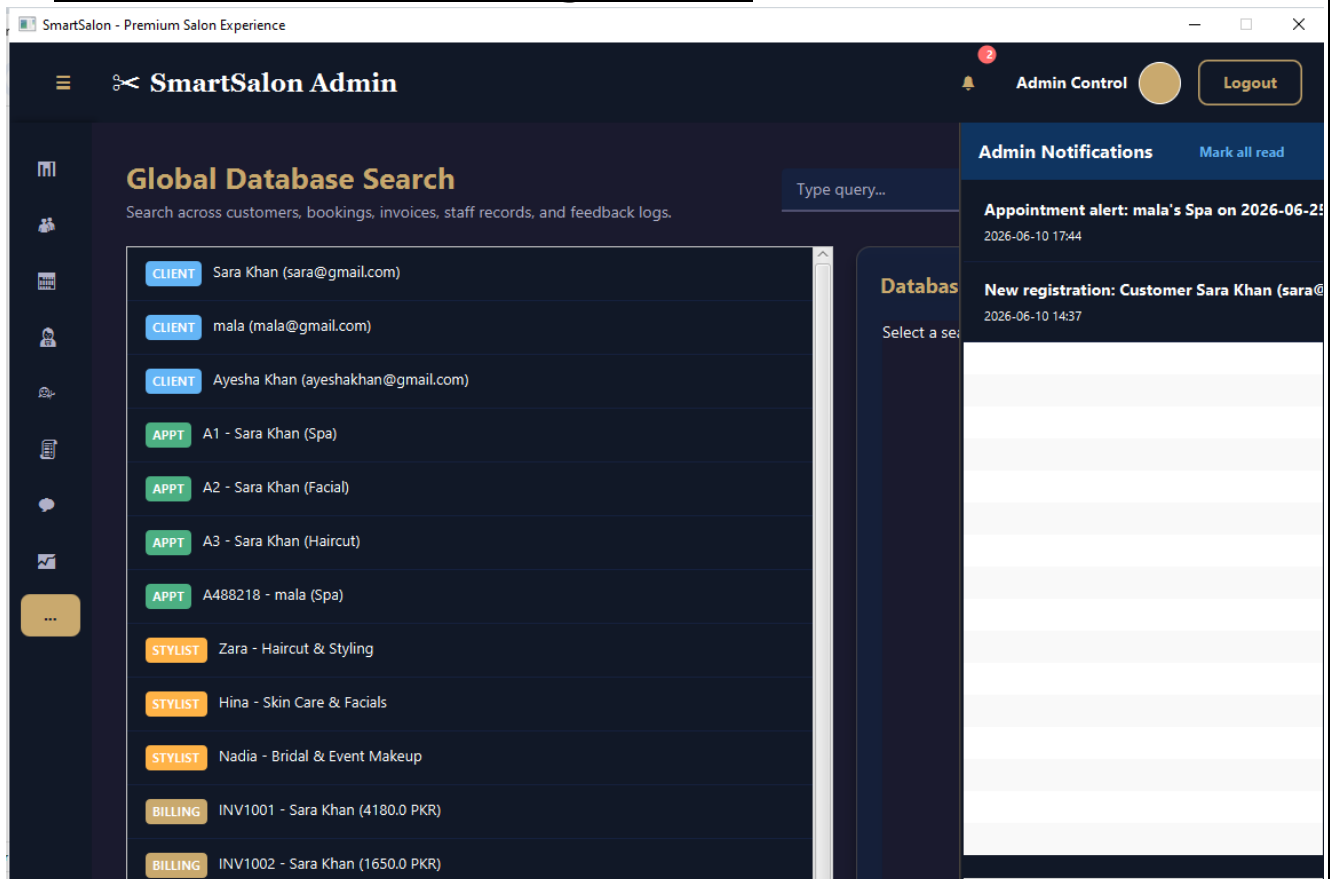
The search interface allows users to query the database across various categories. The search results are as follows:

Category	Item
CLIENT	Sara Khan (sara@gmail.com)
CLIENT	mala (mala@gmail.com)
CLIENT	Ayesha Khan (ayeshakhan@gmail.com)
APPT	A1 - Sara Khan (Spa)
APPT	A2 - Sara Khan (Facial)
APPT	A3 - Sara Khan (Haircut)
APPT	A488218 - mala (Spa)
STYLIST	Zara - Haircut & Styling
STYLIST	Hina - Skin Care & Facials
STYLIST	Nadia - Bridal & Event Makeup
BILLING	INV1001 - Sara Khan (4180.0 PKR)
BILLING	INV1002 - Sara Khan (1650.0 PKR)

Database item details

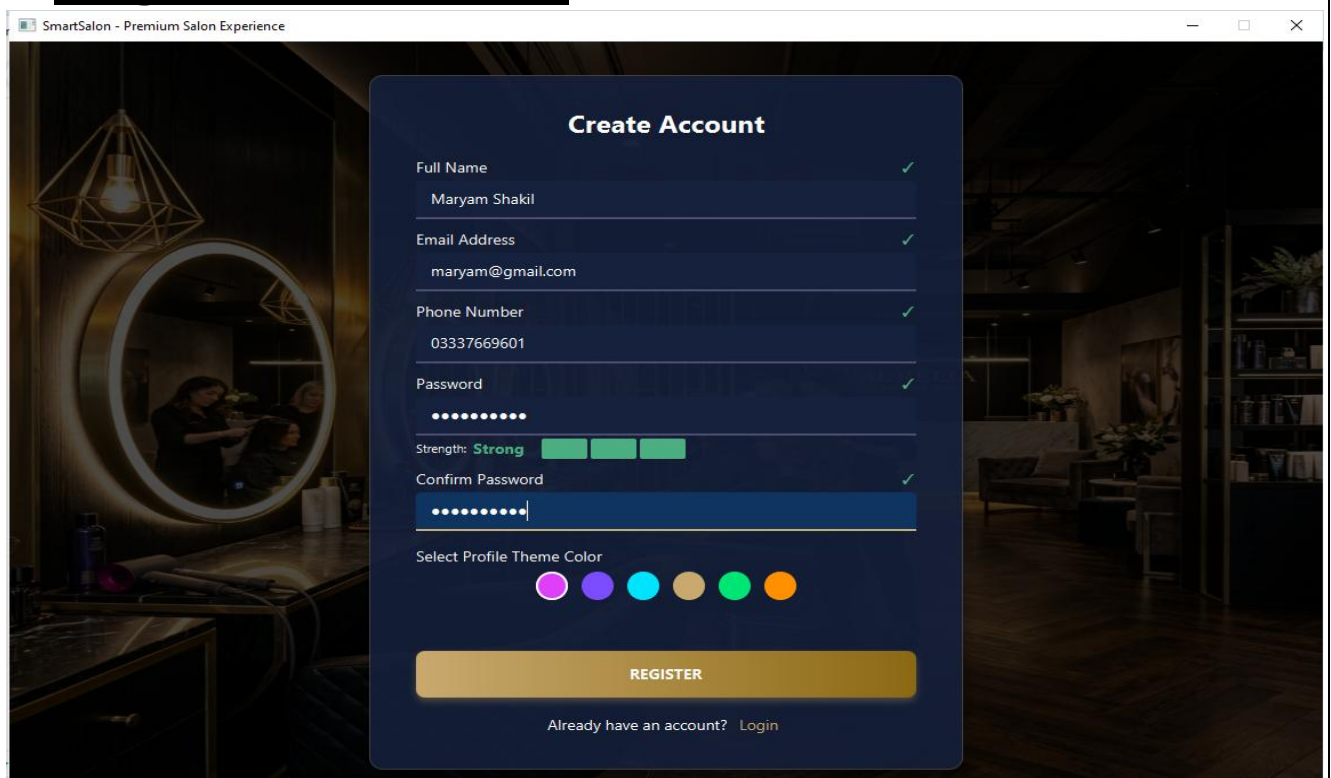
Select a search result item to view details here.

• Notification Management



Customer:

• Register Customer



• Login Customer

SmartSalon - Premium Salon Experience

← Back

Sign In

Customer Admin

Email Address
maryam@gmail.com

Password
••••••••

Forgot Password?

Hint: sara@gmail.com / sara123

LOGIN

Don't have an account? Register Now

• Book Appointments (1.Services)

SmartSalon - Premium Salon Experience

SmartSalon

Maryam Shakil Logout

Dashboard

Book Appointment

My Appointments

My Invoices

Services & Pricing

Feedback & Compl...

My Profile

1. Service → 2. Add-ons → 3. Schedule

Select a Base Service

 Haircut Tailored cut & style 800 PKR	 Facial Herbal glowing skin 1500 PKR
 Makeup Bridal & event glam 2500 PKR	 Spa & Massage Rejuvenating body wash 3000 PKR

Next Step

Booking Summary

Selected Service	Spa
Add-on Upgrades	None
Duration	90 minutes
Scheduled Time	Not scheduled yet
Assigned Stylist	None
Grand Total:	3000.0 PKR

• Book Appointments (2.Add-ons)

SmartSalon - Premium Salon Experience

SmartSalon

Maryam Shakil

Logout

Dashboard

Book Appointment

My Appointments

My Invoices

Services & Pricing

Feedback & Compl...

My Profile

1. Service → 2. Add-ons → 3. Schedule

Choose Add-on Upgrades

Enhance your experience with our luxury add-on services:

☐ **Hair Styling (+500 PKR)**
Blow dry and styling finish with professional products.

☐ **Massage (+800 PKR)**
Head, neck, and shoulder massage (15 mins) for complete relaxation.

☐ **Special Treatment (+1200 PKR)**
Premium conditioning therapy and facial serum upgrade.

Back

Next Step

Booking Summary

Selected Service

Spa

Add-on Upgrades

None

Duration

90 minutes

Scheduled Time

Not scheduled yet

Assigned Stylist

None

Grand Total:

3000.0 PKR

• Book Appointments (3.Schedule)

SmartSalon - Premium Salon Experience

SmartSalon

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Logout

Dashboard

Book Appointment

My Appointments

My Invoices

Services & Pricing

Feedback & Comp...

My Profile

1. Service → 2. Add-ons → 3. Schedule

Schedule Appointment

Assign Stylist
Hina (Skin Care & Facials)

Choose Date
6/24/2026

Select Time Slot

09:00 AM10:00 AM11:00 AM12:00 PM01:00 PM02:00 PM03:00 PM04:00 PM05:00 PM06:00 PM07:00 PM

Special Notes (Optional)

Back

Book Now

Booking Summary

Selected Service

Spa

Add-on Upgrades

None

Duration

90 minutes

Scheduled Time

2026-06-24 @ 04:00 PM

Assigned Stylist

Hina

Grand Total:

3000.0 PKR

✓

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SmartSalon - Premium Salon Experience

SmartSalon

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Maryam Shakil

Logout

Home

Services

Bookings

Profile

...

Services & Pricing Menu

✂️

Haircut & Styling

Premium haircuts, styling, hair washes, and conditioning tailored for you...

Price: 800 PKR • Duration: 30m

Book Now

💆‍♀️

Skin Facials

Natural herbal facials, deep cleansing, blackhead extraction, and skin ...

Price: 1500 PKR • Duration: 45m

Book Now

💄

Event Makeup

Luxury makeup application for brides, parties, catalog shoots, and speci...

Price: 2500 PKR • Duration: 60m

Book Now

💆‍♂️

Relaxing Spa & Massage

Full body deep tissue massages, aromatherapy, and hot stone muscle ...

Price: 3000 PKR • Duration: 90m

Book Now

Premium Add-on Menu (Chainable)

- **Submits Feedbacks and Complaints**

SmartSalon - Premium Salon Experience

X

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SmartSalon

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Logout

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Home

Calendar

Services

Bookings

Feedback & Complaints

Profile

Submit Feedback & Complaints

We value your feedback! Rate your recent experience and let us know how we can improve.

Overall Rating

★ ★ ★ ★ ★

Feedback Category

Select category ▼

Comments

Please describe your experience in detail (minimum 20 characters)...

0/500 characters (min 20)

SUBMIT FEEDBACK

- **My Profile Settings**

SmartSalon - Premium Salon Experience

Maryam ShakilLogout

HomeDashboardProfile SettingsAppointmentsCalendarReportsUsersSettingsHelp

My Profile Settings

Personal Details

Choose Profile Theme Color

Full Name

Maryam Shakil

Email Address (Locked)

maryam@gmail.com

Phone Number

03337669601

Change Password

Current Password

Enter current password

New Password

Enter new password (min 6 characters)

Strength: Weak

Confirm New Password

Re-enter new password

SAVE ALL CHANGES

7. Functional Requirements

Customer Module

Customers can:

- Register new accounts.
- Login securely.
- Reset forgotten passwords.
- Book appointments.
- View appointment history.
- Cancel appointments.
- View invoices.
- Browse services.
- Submit feedback.
- Update profile information.

Admin Module

Administrators can:

- View dashboard statistics.
 - Manage customers.
 - Manage appointments.
 - Manage staff members.
 - Manage salon services.
 - Manage invoices.
 - Review customer complaints.
 - Generate reports and analytics.
 - Perform global searches.
-

9. Key Features

Customer Features

- Appointment Booking
- Service Selection
- Invoice Viewing
- Feedback Submission
- Profile Management

Admin Features

- Customer Management
- Staff Management
- Appointment Scheduling
- Invoice Generation
- Complaint Handling
- Revenue Analysis
- Business Reporting

10. Results

The Smart Salon Management System successfully automates salon operations and provides an efficient platform for both customers and administrators. The implementation of Factory, Decorator, Observer, and Singleton design patterns improves software quality, flexibility, and maintainability.

11. Conclusion

The Smart Salon Management System is a complete desktop application developed using Java and JavaFX. The project demonstrates practical implementation of Object-Oriented Programming concepts, MVC architecture, and software design patterns. The system effectively manages salon activities while providing a professional and user-friendly experience.
